

The Moon Race: USSR vs US

Intelligent Agents — Exercise 03
University of Haifa

Authors

Renat Karimov
Alon Engel

Group code: `anrbj666`

Course: Intelligent Agents
Course instructor: Dr. Yoram Segal

June 12, 2026

Contents

1	Cold War Context and the Space Race Begins	2
1.1	Superpower rivalry after 1945	2
1.2	Rockets, missiles, and prestige	2
2	Soviet Early Lead: Sputnik and Vostok	4
2.1	Sputnik and the shock in America	4
2.2	First human in space: Yuri Gagarin	4
3	American Response: Mercury, Gemini, Apollo	6
3.1	NASA and the crash program	6
3.2	From Earth orbit to the lunar landing goal	6
4	Key Missions and Turning Points	8
4.1	Robotic Luna, Ranger, and Surveyor	8
4.2	Apollo 11 and lunar samples	8
5	Propaganda, Politics, and Public Opinion	10
5.1	Media and ideological messaging	10
5.2	Cost, risk, and congressional support	10
6	Legacy: Who Won and What Remains	12
6.1	End of Apollo and later cooperation	12
6.2	Lessons for science and exploration	12
A	How This Book Was Made	14
A.1	The agent pipeline	14
A.2	From IPC orchestration to a CrewAI crew	14
A.3	Observability and cost	14

1 Cold War Context and the Space Race Begins

1.1 Superpower rivalry after 1945

After the Second World War, the United States and the Soviet Union emerged as rival superpowers whose competition shaped global politics for decades. Their contest was ideological, military, and economic, but it was also deeply symbolic: each side sought visible proof that its system could out-build and out-think the other. Space, still an untouched frontier, offered exactly that kind of stage.

The rivalry hardened quickly. Berlin, Korea, and the nuclear arms race left little room for trust, and reliable intelligence on the other side was scarce and prized. In that climate, technological milestones doubled as strategic signals: a successful launch told allies and adversaries alike that a nation could deliver a warhead across continents, so the line between civilian achievement and military deterrence was deliberately blurred.

Prestige, then, was never a side effect of the space race; it was a central goal. Governments invested enormous resources because being first carried weight in capitals, newspapers, and living rooms around the world. The Moon eventually became the ultimate prize precisely because it was difficult, expensive, and visible to everyone who looked up at night. [6] [1]

1.2 Rockets, missiles, and prestige

The rockets that carried satellites and cosmonauts were direct descendants of wartime weapons. Germany's V-2, the first large liquid-fueled ballistic missile, proved that controlled rocket flight at the edge of space was possible, and at the war's end both superpowers raced to recruit its engineers and seize its hardware. The expertise gathered in 1945 became the seed of two parallel national programs.

Through the 1950s, missile development and space ambition advanced together. The same boosters designed to deliver nuclear warheads could, with modest changes, loft scientific payloads, so investment in one capability quietly strengthened the other. This dual use meant that every rocket test was watched closely abroad, and that failures were as politically costly as they were technically instructive.

Prestige tied the whole enterprise together. A reliable, powerful rocket signaled industrial maturity, scientific depth, and military credibility in a single stroke. Because these signals were read by allies, adversaries, and domestic audiences at once, governments were willing to fund ambitious programs whose payoff was measured as much in influence as in engineering. [2] [4]

Table 1: Selected Moon Race milestones

Year	Program	Milestone
1957	Sputnik	First artificial satellite
1961	Vostok	First human orbital flight
1965	Gemini	Rendezvous and EVA practice
1969	Apollo 11	First crewed lunar landing
1975	Apollo-Soyuz	Post-race cooperation

סיכום הפרק

פרק זה מציג את רקע המלחמה הקרה ואת ראשית מרוץ החלל. שתי מעצמות-העל הפכו טילים בליסטיים להישג יוקרתי שנועד לתעמולה ולהוכחת עליונות טכנולוגית.



Figure 1: Apollo 11 Saturn V on Launch Pad 39A before the Moon landing mission.

2 Soviet Early Lead: Sputnik and Vostok

2.1 Sputnik and the shock in America

On 4 October 1957 the Soviet Union launched Sputnik 1, the first artificial satellite, and its simple radio beep, audible to amateur listeners worldwide, announced that the space age had begun. The achievement was modest in mass but enormous in meaning: a Soviet object was now passing over American territory several times a day, beyond any power to stop it.

In the United States the reaction was swift and anxious. Commentators spoke of a Sputnik crisis, questioning whether American science and education had fallen behind. The launch re-framed space as a domain of national security and pride, and the political pressure it generated would soon reshape budgets and institutions.

The most lasting response was structural. Within a year the United States created NASA to lead civilian spaceflight and expanded federal support for science and engineering education. Sputnik thus did more than orbit the Earth; it set in motion the institutional machinery that would eventually carry Americans to the Moon. [1] [5]

2.2 First human in space: Yuri Gagarin

On 12 April 1961 Yuri Gagarin orbited the Earth aboard Vostok 1, becoming the first human in space and handing the Soviet Union another decisive first. The single orbit lasted under two hours, but its symbolic weight was immense: a human being had left the planet and returned, and a Soviet citizen had done it first.

The flight was the product of careful, secretive engineering led by Sergei Korolev, whose identity the Soviet state concealed for security reasons. Vostok's automated systems, life support, and reentry capsule represented years of incremental, high-risk testing, much of it hidden from the public until success could be announced.

Gagarin's flight intensified the contest. Coming only weeks before the first American sub-orbital flight, it underscored how far ahead the Soviet program appeared and pushed Washington to seek a goal dramatic enough to change the narrative. That goal would soon be named: the Moon. [6] [2]

סיכום הפרק

פרק זה עוסק ביתרון הסובייטי המוקדם: שיגור הלוויין הראשון והטסת האדם הראשון לחלל. הישגים אלו זעזעו את ארצות הברית והאיצו את תגובתה.

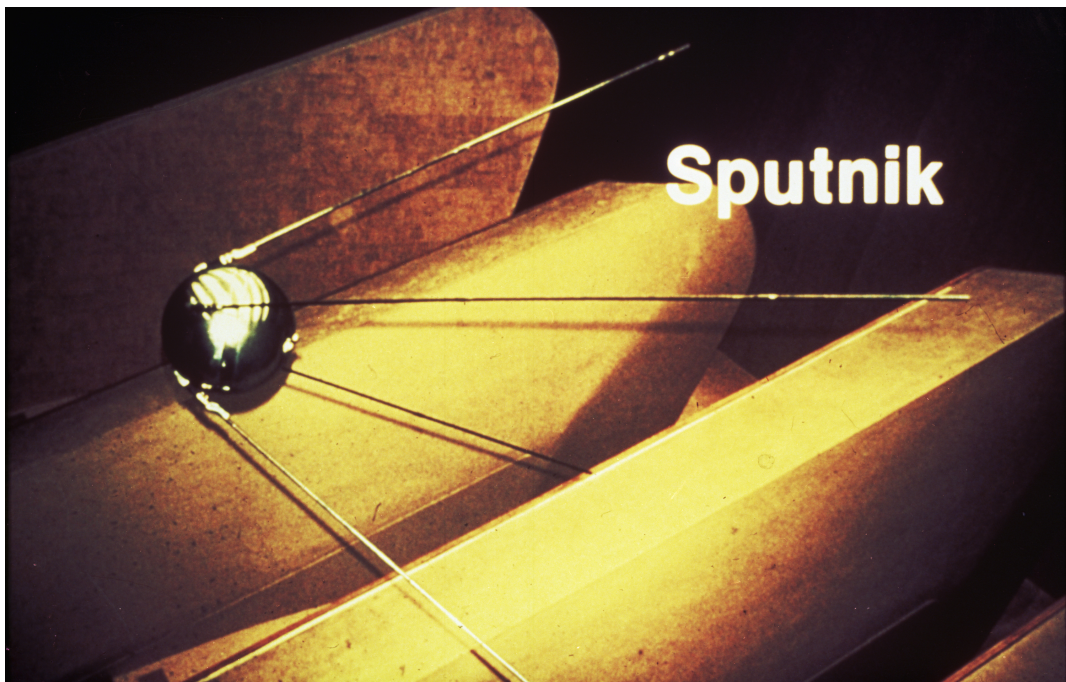


Figure 2: Sputnik era - the Soviet Union launched the first artificial satellite in 1957.

3 American Response: Mercury, Gemini, Apollo

3.1 NASA and the crash program

Faced with repeated Soviet firsts, the United States organized an unprecedented peacetime mobilization of engineering talent. NASA grew rapidly, contracting with universities and thousands of companies to build launch vehicles, spacecraft, tracking networks, and test facilities almost simultaneously. The result was a crash program whose scale rivaled the largest industrial efforts in the nation's history.

Project Mercury came first, proving that Americans could survive launch, orbit, and reentry. Its successes were hard-won and public, and they restored a measure of confidence even as they revealed how much remained unknown about living and working in space.

Managing such a program demanded new methods. Systems engineering, rigorous testing schedules, and tight configuration control became as important as any single rocket, because coordinating so many contractors under deadline pressure was itself a formidable technical challenge. [4] [5]

3.2 From Earth orbit to the lunar landing goal

In May 1961 President Kennedy committed the United States to landing a man on the Moon and returning him safely before the decade's end. The pledge was audacious: the country had barely fifteen minutes of human spaceflight experience, yet it now had a deadline and a destination that the whole world could understand.

Choosing the Moon was a strategic decision as much as a scientific one. It was far enough ahead that the Soviet lead in early milestones would not decide the outcome, and dramatic enough that success would be unmistakable. The goal concentrated national effort on a single, measurable objective.

Project Gemini bridged the gap between ambition and capability. Its flights practiced the rendezvous, docking, long-duration flight, and spacewalking that a lunar mission would require, turning Kennedy's promise into a sequence of solvable engineering problems. [4] [1]

The ideal rocket equation relates velocity change to mass ratio:

$$\Delta v = v_e \ln \left(\frac{m_0}{m_f} \right) \quad (1)$$

where v_e is effective exhaust velocity, m_0 initial mass, and m_f final mass.

סיכום הפרק

פרק זה מתאר את תוכניות החלל האמריקאיות - מרקורי, ג'מיני ואפולו - שריכזו משאבים לאומיים כדי להשיג נחיתה על הירח לפני ברית המועצות.

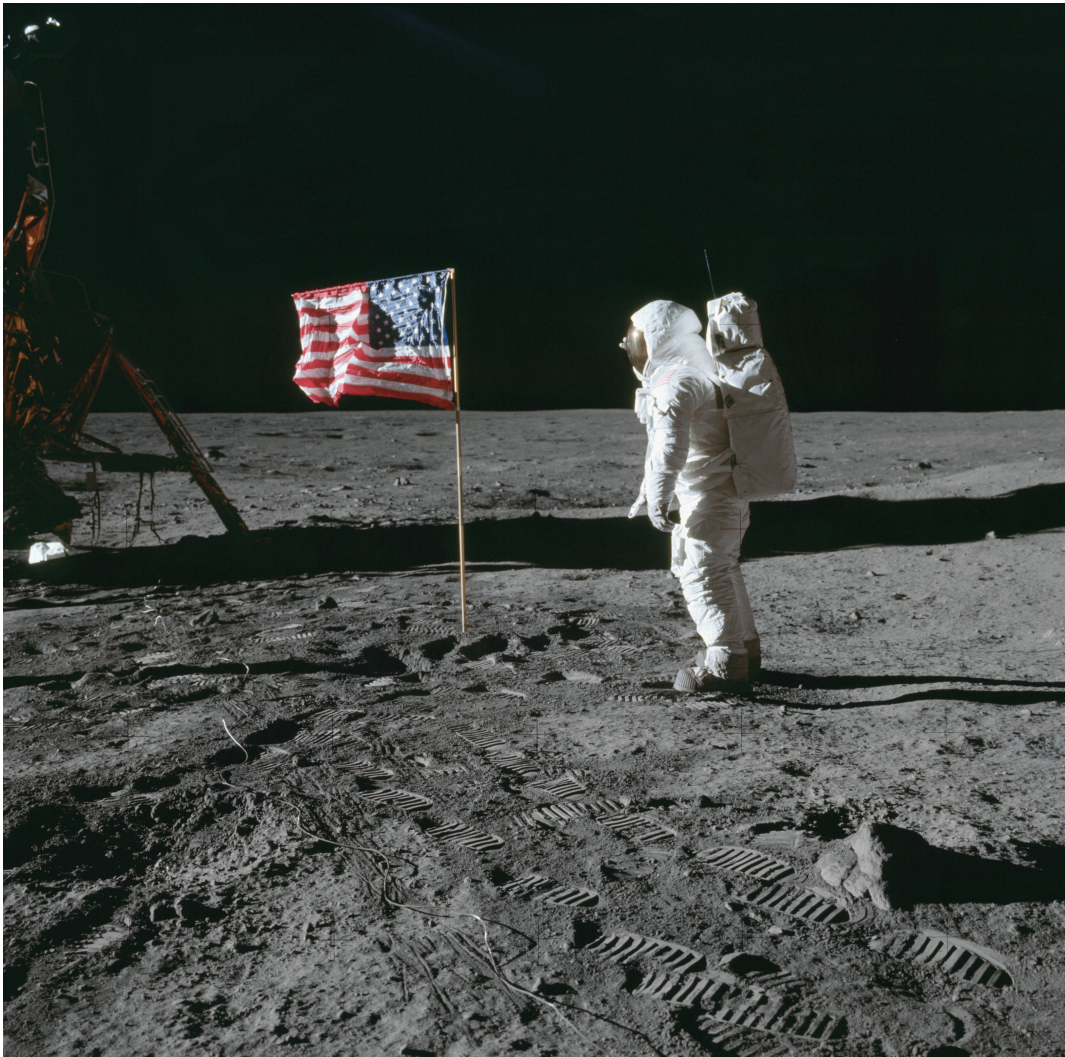


Figure 3: Buzz Aldrin on the Moon during Apollo 11.

4 Key Missions and Turning Points

4.1 Robotic Luna, Ranger, and Surveyor

Before any crew could land, both nations sent robotic scouts to study the Moon. The Soviet Luna program achieved a string of firsts, including the first impact on the lunar surface and the first images of its far side, expanding humanity's map of a world no one had yet touched.

American robotic missions followed a deliberate progression. Ranger probes returned close-up images as they crashed into the surface, while the Surveyor landers demonstrated soft landing and showed that the lunar soil could bear the weight of a spacecraft, easing fears that a lander might simply sink.

Together these missions reduced risk. They mapped hazards, measured surface properties, and validated the navigation and landing techniques that crews would later depend on, turning the Moon from an unknown into a surveyed destination. [6] [5]

4.2 Apollo 11 and lunar samples

On 20 July 1969 Apollo 11's lunar module Eagle touched down in the Sea of Tranquility, and Neil Armstrong and Buzz Aldrin became the first humans to walk on another world. Hundreds of millions watched or listened as a goal set eight years earlier was fulfilled with months to spare.

The crew did more than plant a flag. They deployed scientific instruments and collected about twenty-two kilograms of rock and soil, the first material ever returned from the Moon, which scientists would study for decades to understand the Moon's age and origin.

Apollo 11 marked the decisive moment of the race. The United States had achieved the most visible objective first, and while exploration continued, the central question of who would reach the Moon had been answered. [5] [4]

סיכום הפרק

פרק זה סוקר משימות מפתח: גשושיות רובוטיות שמיפו את הירח, ולאחריהן נחיתת אפולו 11 שהביאה דגימות ראשונות מפני הירח.

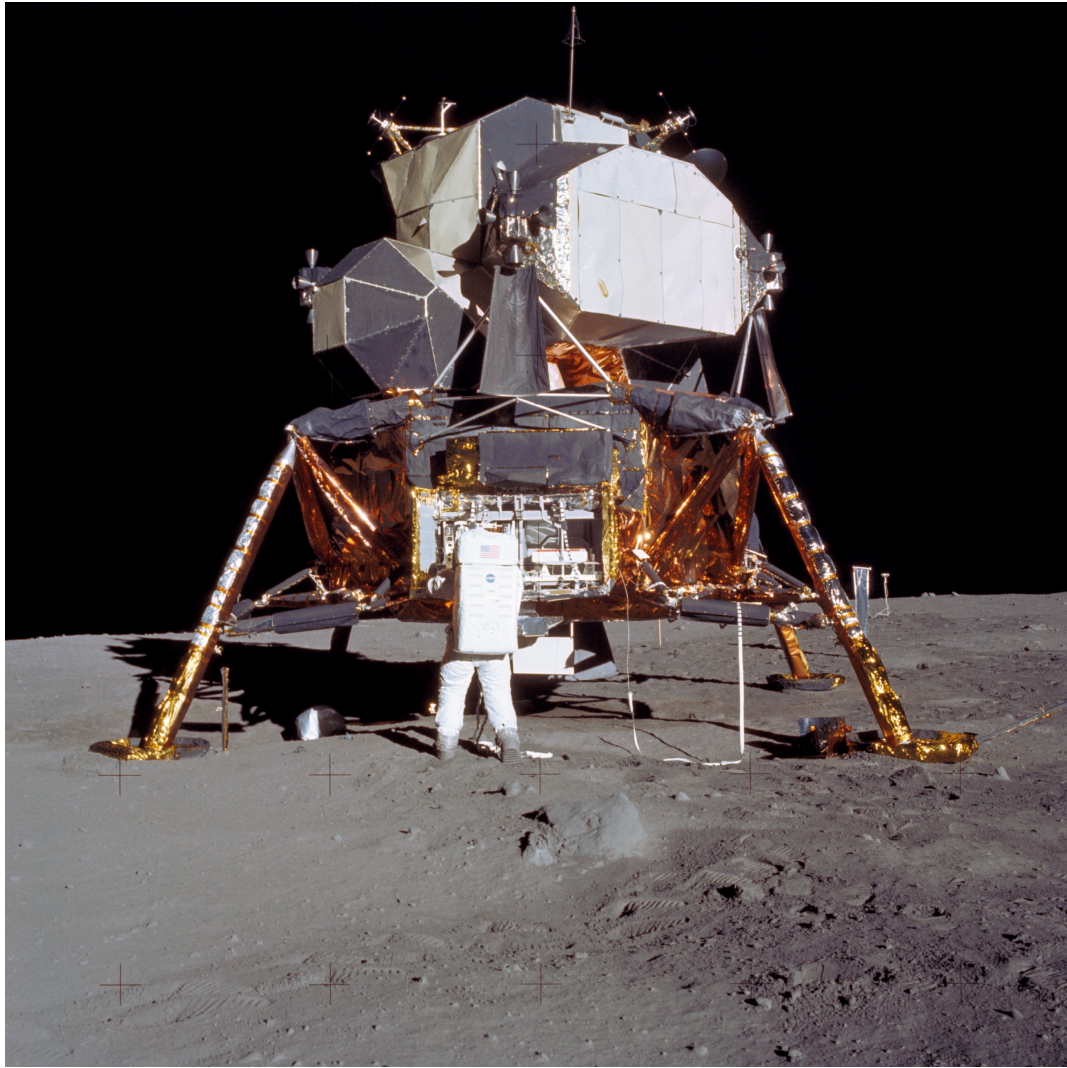


Figure 4: Apollo 11 Lunar Module Eagle on the Moon surface.

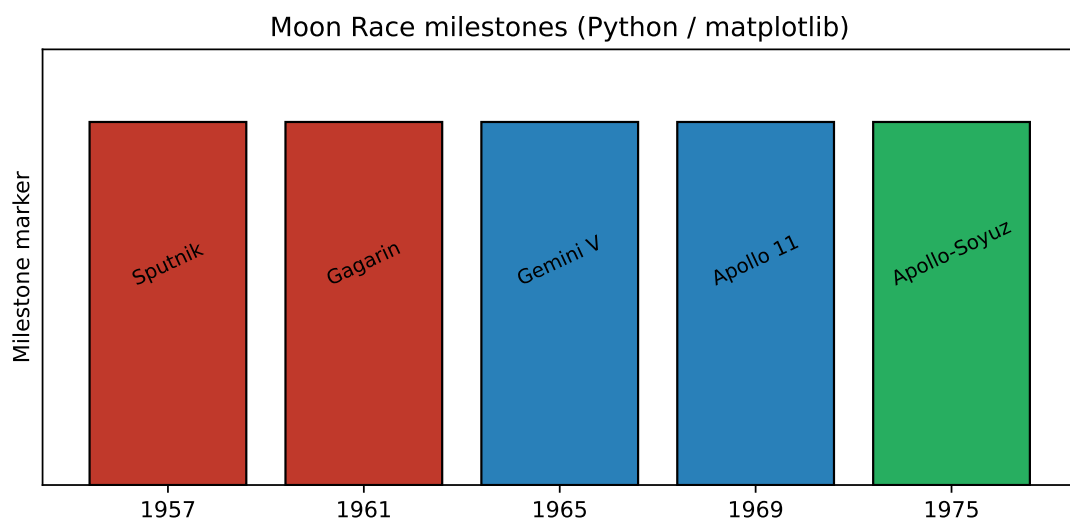


Figure 5: Moon Race milestones chart generated with Python (matplotlib).

5 Propaganda, Politics, and Public Opinion

5.1 Media and ideological messaging

Spaceflight was always partly a performance. Both superpowers presented their achievements as evidence that their political and economic systems were superior, and each launch was packaged for domestic audiences and the wider world as proof of progress.

Their strategies differed sharply. The Soviet Union tended to announce successes only after they were secure, keeping failures hidden, while the United States increasingly broadcast its missions live, accepting the risk of public failure in exchange for the credibility of openness.

The imagery endured. Photographs such as Earthrise, taken from lunar orbit, transcended propaganda and reshaped how people everywhere saw their own planet, showing that the race produced cultural touchstones as well as political points. [1] [6]

5.2 Cost, risk, and congressional support

Ambition came at a steep price. At its peak the Apollo program consumed a significant share of the federal budget and employed hundreds of thousands of people, and sustaining that investment required continual political justification.

Risk was equally real. The 1967 Apollo 1 fire, which killed three astronauts during a ground test, forced a sweeping reassessment of design and safety and reminded everyone that the race was conducted at the edge of what was survivable.

Public and congressional support proved finite. As early goals were met and costs mounted, enthusiasm cooled, and the same political forces that had launched the program began to question how long it could continue at such expense. [4] [2]

סיכום הפרק

פרק זה בוחן תעמולה, פוליטיקה ודעת קהל. שני הצדדים הציגו את הישגי החלל כראיה לעליונות שיטתם, תחת לחצי תקציב וסיכון.



Figure 6: Earthrise from Apollo 8 - icon of public opinion and the space race.

6 Legacy: Who Won and What Remains

6.1 End of Apollo and later cooperation

After Apollo 11, the United States flew several more lunar missions before ending the program in 1972, as budgets tightened and priorities shifted. The Soviet crewed lunar effort, hampered by the failures of its giant N1 rocket and the death of Korolev, never reached the Moon at all.

Competition gradually gave way to contact. In 1975 the Apollo-Soyuz Test Project docked American and Soviet spacecraft in orbit, a symbolic handshake that signaled the race's most intense phase was over.

That thaw set a precedent. The cooperation rehearsed in the 1970s would later mature into joint programs, culminating in the International Space Station, where former rivals share a single laboratory in orbit. [5] [1]

6.2 Lessons for science and exploration

The Moon race left a deep scientific legacy. The returned samples, instruments, and data transformed lunar science, while the technologies developed for spaceflight found uses far beyond it, from materials and computing to global communications.

It also offered organizational lessons. Achieving the landing required disciplined systems engineering, relentless testing, and clear goals shared across enormous teams, a model later studied by engineers and managers in many fields.

Perhaps the most durable lesson is about motivation and its limits. A single dramatic goal can mobilize extraordinary effort, but sustaining exploration over the long term depends on broader purposes, an insight that still informs how nations plan their journeys beyond Earth. [6] [4]

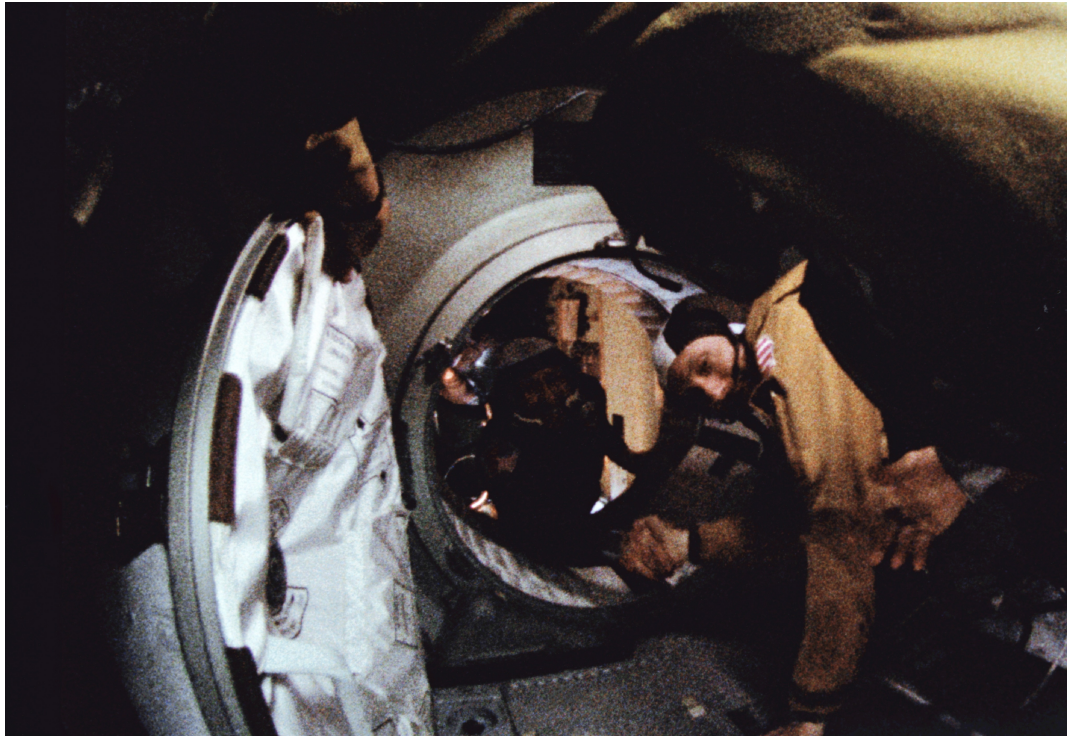


Figure 7: Apollo-Soyuz handshake in space - beginning of post-race cooperation.

סיכום הפרק

פרק זה מסכם את המורשת: סיום עידן אפולו והמעבר מתחרות לשיתוף פעולה בין-לאומי - מאפולו-סויוז ועד תחנת החלל הבין-לאומית.

A How This Book Was Made

This appendix documents the engineering behind the book itself: the CrewAI agent pipeline that produced the content and the LaTeX toolchain that compiled it. It is included to demonstrate the architecture, not merely describe it [3].

A.1 The agent pipeline

The book is generated by a sequential CrewAI crew. Each agent maps to a job title in a small publishing house, and the output of one stage flows as context into the next. The three language-model stages route every call through a central API gatekeeper that enforces rate limits, retries, and cost logging.

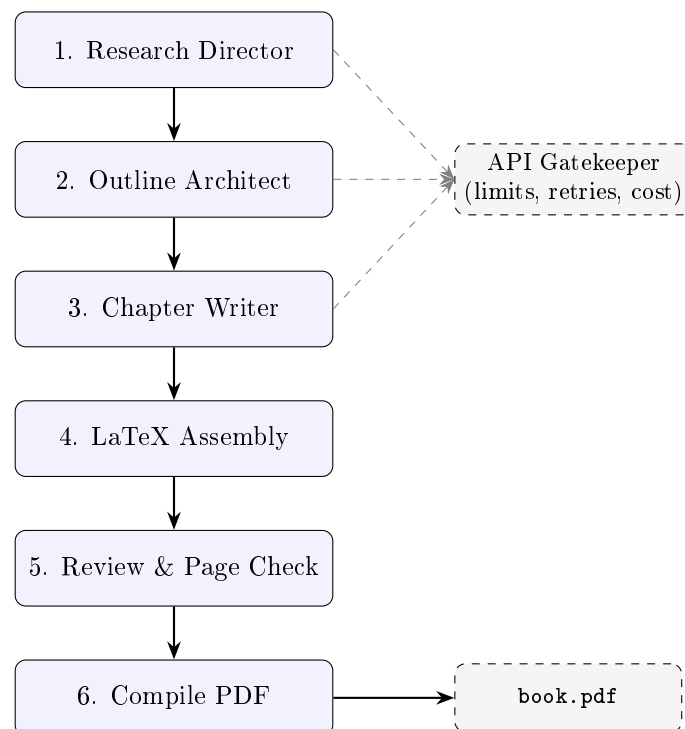


Figure 8: CrewAI book-generation pipeline. Solid arrows show the sequential hand-off of artifacts; dashed arrows show language-model calls routed through the gatekeeper.

A.2 From IPC orchestration to a CrewAI crew

An earlier exercise orchestrated agents directly with inter-process communication, wiring every queue, retry, and hand-off by hand. That approach is flexible and fast for a proof of concept, but it pushes all coordination onto the developer. CrewAI moves that boilerplate into a framework: roles, tasks, and a process type replace custom message plumbing, so the team can focus on prompts, schemas, and quality gates. The trade-off is less low-level control in exchange for clearer structure and easier hand-over.

A.3 Observability and cost

Because a production agent is a system rather than a single prompt, every language-model call is logged as a JSON line with token counts and an estimated cost, and the gatekeeper records any request that exceeds a configured budget. This makes a run reproducible and auditable: the

same configuration yields the same book, and the cost of a live run can be reported from the logs rather than guessed.

References

- [1] Deborah Cadbury. *Space Race: The Epic Battle Between America and the Soviet Union for Dominion of Space*. HarperCollins, 2006.
- [2] James Harford. *Korolev: How One Man Masterminded the Soviet Drive to Beat America to the Moon*. John Wiley and Sons, 1997.
- [3] Renat Karimov and Alon Engel. Crewai publishing pipeline for exercise 03. 2026. Team analysis of agent-generated book production.
- [4] John M. Logsdon. *John F. Kennedy and the Race to the Moon*. Palgrave Macmillan, 2010.
- [5] NASA. Apollo 11 mission overview. NASA History Office, 2019.
- [6] Asif A. Siddiqi. *Challenge to Apollo: The Soviet Union and the Space Race, 1945–1974*. NASA History Division, 2010.